

REGULAR ARTICLE

Analgesia by cooling vibration during venipuncture in children with cognitive impairment

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ABSTRACT

Aim: Children with cognitive impairment experience pain more frequently than healthy children and are more likely to require venipuncture or intravenous cannulation for various procedures. They are frequently unable to report pain and often receive poor pain assessment and management. This study assessed the effectiveness of physical analgesia during vascular access in children with cognitive impairments.

Methods: We conducted a prospective randomised controlled study at a tertiary-level children's hospital in Italy from April to May 2015 to assess whether a cooling vibration device called Buzzy decreased pain during venipuncture and intravenous cannulation in children with cognitive impairment. None of the children had verbal skills and the main cognitive impairments were cerebral palsy, epileptic encephalopathy and genetic syndromes.

Results: We tested 70 children with a median age of nine years: 34 in the Buzzy group and 36 in the no-intervention group. Parents were trained in the use of the Noncommunicating Children's Pain Checklist – postoperative version scale, and they reported no or mild procedural pain in 32 cases (91.4%) in the Buzzy group and in 22 cases (61.1%) in the no-intervention group ($p = 0.003$).

Conclusion: Cooling vibration analgesia during vascular access reduced pain in children with cognitive impairment.

INTRODUCTION

Needle-related procedures are the most common source of pain and distress experienced by children when they are in hospital (1). Children with cognitive impairment from any cause experience pain more frequently than healthy children (2) and are more likely to require venipuncture or intravenous cannulation for diagnostic or therapeutic procedures. Pain in this population often goes unrecognised because these patients are frequently unable to report their pain (3). Evidence shows that children with cognitive impairment often receive poor pain assessment and management compared to the normal population (4–6), despite the availability of specific pain assessment tools (7).

Pain control is recommended during needle-related procedures in children (8). Providing a comfortable envi-

ronment and nonpharmacological techniques are the first steps in pain management, while the application of local anaesthetics is the most used pharmacological approach.

The application of distraction techniques has not been extensively studied in children with cognitive impairment, and their use may be limited by the severity of the disability. Using physical methods that employ cold temperatures or

Key notes

- Children with cognitive impairment experience pain more frequently than healthy children and are more likely to require venipuncture or intravenous cannulation for various procedures.
- We carried out a randomised controlled trial to assess the analgesic efficacy of a cooling vibration device called Buzzy.
- Parents reported no or mild procedural pain in 91.4% of the children in the Buzzy group and in 61.1% of the children in the no-intervention group.

Abbreviations

IQR, Interquartile range; NCCPC-PV, Noncommunicating Children's Pain Checklist – postoperative version; NCCPC-R, Noncommunicating Children's Pain Checklist – revised general version.

vibration is effective in providing pain relief during venipuncture (9,10).

Physical methods are based on the gate control theory (11), which states that nociception from the peripheral to central nervous system is modulated by a gate system in the dorsal horn of the spinal cord. Stimulating touch and temperature receptors decreases a subsequent painful sensation.

Buzzy (MMJ Labs, Atlanta, Georgia, USA) is a device that combines a cold temperature and high frequency vibration. It is a reusable $8 \times 5 \times 2.5$ cm plastic bee that contains a battery-operated vibrating motor and a mechanism to attach an ice pack underneath.

Buzzy was initially found to be effective in pain relief during intravenous cannulation in adults (12). After that, several studies demonstrated its effectiveness in providing pain relief in the paediatric population during venipuncture, intravenous cannulation and painful injections (13–17).

There have been no studies to investigate the effect of the Buzzy device in children with cognitive impairment.

The application of topical anaesthesia is standard of care for children with and without cognitive impairment in our Institute, although this is not always possible due to the parents' refusal, time constraints and known hypersensitivity to the drug.

The purpose of this study was to test the effectiveness of the Buzzy device in providing pain relief during venipuncture or intravenous cannulation in children with cognitive impairment.

PATIENTS AND METHODS

This study was an open randomised controlled trial with two parallel arms. It was conducted between April and May 2015 in the paediatric and surgical ward and emergency department of the Institute for Maternal and Child Health IRCCS Burlo Garofolo of Trieste, Italy, a tertiary-level children hospital.

The study received approval from the Independent Bioethic Committee of the Institute. All the children's parents provided written, informed consent before participation. The trial was registered with ClinicalTrials.gov, number (NCT02434731).

The eligible subjects were children from four to 17 years of age, with cognitive impairment and no verbal communication, who needed venipuncture or intravenous cannulation for diagnosis or treatment, without the application of topical anaesthetic due to the parents' refusal and/or time constraints.

The exclusion criteria were as follows: critically ill children; the presence of abrasion, infected or damaged skin in the area where the device would be placed; the use of topical, enteral or parenteral analgesics within eight hours of presentation; known nerve damage or peripheral neuropathy; children affected by a disease associated with cold hypersensitivity like Raynaud's disease or sickle cell anaemia and a diagnosis of Prader–Willi syndrome, which is characterised by low sensitivity to painful stimuli.

The eligible children were randomly allocated to receive either the application of the Buzzy device and frozen ice pack or no intervention. When the Buzzy was applied it was put about 5 cm above the needle procedure area and switched on for at least 15 seconds before venipuncture. The vibration was continued until the end of the procedure.

A randomisation list with a fixed block size of ten was generated using a computer-based method by an independent epidemiologist at the Clinical Epidemiology and Health Services Research Unit of the Institute for Maternal and Child Health IRCCS Burlo Garofolo, Trieste, Italy.

Allocation concealment was guaranteed through the use of closed opaque envelopes, consecutively numbered, each containing the allocation group. After obtaining each informed consent, the nurse opened the lowest numbered envelope available and assigned the patient to the corresponding group.

The primary study outcome was procedural pain score in the two groups, recorded by parents using the Noncommunicating Children's Pain Checklist – postoperative version (NCCPC-PV) scale.

A trained nurse instructed parents about the use of the NCCPC-PV scale, which is validated for children aged 3–18 years unable to verbalise because of mental or intellectual impairments or disabilities (6,18,19). Any person involved in the child's care may use the scale.

From the original checklist two versions have been derived: the postoperative version (NCCPC-PV) and a revised general version (NCCPC-R). The difference between the two versions is that the former does not include one of the checklist items, the eating and sleeping subscale. The postoperative version was the most suitable for this study (18,19).

NCCPC-PV score ranges from 0 to 80. A pain score ≤ 10.0 indicates absence of pain or mild pain; a score > 10.0 indicates the presence of moderate-to-severe pain (6).

The secondary outcomes were the needle procedure success rate at the first attempt in the two groups and the number and type of adverse events in the two groups.

The following variables were collected for each child: age; gender; weight; the presence of fever or dehydration; the presence of pain before the procedure; diagnosis; type and site of procedure and the experience of the staff performing the procedure, based on the number of procedures performed in the previous month. The operator performing the procedure filled out the data collection form.

Statistical analyses

As no data were available in the literature concerning the use of the Buzzy device in children with cognitive impairment, we based our sample size calculation on preliminary data collected at our Institute. Hypothesising on the basis of these data, a mean NCCPC-PV score of 5.0 in the Buzzy group versus 10.0 in the no-intervention group (with a standard deviation of 7.0), 70 children – 35 for each group – were needed to carry out the study, with $\alpha = 0.05$ and $\beta = 0.20$.

Analyses were carried out according to the intention-to-treat principle. Continuous variables are reported as median and interquartile range (IQR), with categorical data presented as numbers and percentages. Differences between the groups were evaluated with the Mann–Whitney *U*-test for continuous variables or with the chi-square test or Fisher's exact test for categorical variables. Data were analysed with SPSS software, version 21.0 (IBM Corp., Armonk, NY, USA). A *p* value of <0.05 was considered statistically significant.

RESULTS

We assessed 92 children for eligibility and, of these, nine were excluded and 12 declined to participate. A total of 71 children entered the study: 35 were randomised to the Buzzy group and 36 to the control group (Fig. 1). One child randomised to the Buzzy group received no intervention. Following the intention-to-treat principle, this subject was considered to belong to his respective randomisation group. No subject was lost to follow-up.

The baseline characteristics of the two groups were similar (Table 1). None of the children were capable of verbal communication, due to their cognitive impairment. This was due to cerebral palsy in 28 of the children (39.4%), epileptic encephalopathy in 18 (25.3%), genetic syndromes in 13 (18.4%) and other different causes in 12 (16.9%).

Table 2 shows the study results. The NCCPC-PV scores evaluated during the procedure were statistically significantly lower in the Buzzy group than the no-intervention group (median 3.0, IQR 0.0–7.0 and 8.0, IQR 0.3–17.0, respectively, *p* = 0.03) (Fig. 2). Children reported no or mild pain in 32 cases (91.4%) in the Buzzy group and in 22 cases (61.1%) in the no-intervention group (*p* = 0.003).

There was no statistically significant difference in the procedural success rate between the two groups: most procedures were successful at the first attempt in both groups, 90.6% in the Buzzy group versus 87.1% in the no

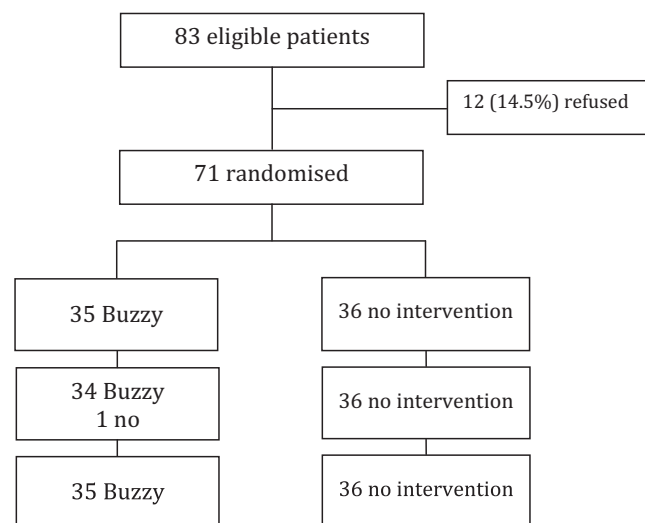


Figure 1 Study algorithm (intention-to-treat analysis).

Table 1 Patients characteristics

Characteristics	Buzzy group (n = 35)	No-intervention group (n = 36)
Age in years, median (IQR)	9.0 (8.0–12.0)	9.0 (7.3–12.0)
Male sex %	19 (54.3)	23 (63.9)
Weight in kilograms, median (IQR)	23.0 (17.0–40.0)	27.5 (21.5–42.8)
Presence of fever, number (%)	1 (2.9)	1 (2.8)
Presence of dehydration, number (%)	2 (5.7)	2 (5.6)
Diagnosis:		
Cerebral palsy	13 (37.1)	15 (41.7)
Epileptic encephalopathy	10 (28.6)	8 (22.2)
Genetic syndromes	7 (20.0)	6 (16.7)
Other	5 (14.3)	7 (19.4)
Procedure type, number (%)		
Venipuncture	28 (80.0)	28 (77.8)
Intravenous cannulation	7 (20.0)	8 (22.2)
Site of the procedure, number (%)		
Arm	21 (60.0)	20 (55.5)
Hand	14 (40.0)	16 (44.5)
Procedure performed by the operator in the previous month, number (%)		
5–10	1 (2.9)	0
>10	34 (97.1)	36 (100%)

Table 2 Main study outcomes

Main Outcomes	Buzzy group (n = 35)	No-intervention group (n = 36)	<i>p</i> value
NCCPC-PV during procedure, median (IQR)	3.0 (0.0–7.0)	8.0 (0.3–17.0)	0.03
NCCPC-PV during procedure in categories, number (%)			
No pain or mild pain (score ≤10)	32 (91.4)	22 (61.1)	0.003
Moderate-to-severe pain (score >10)	3 (8.6)	14 (38.9)	
Procedural success rate at the first attempt, number (%)	31 (88.6)	32 (88.9)	1.0*
Adverse events, number (%)	3 (8.6%)	1 (2.8)	0.36*

*Fisher's exact test.

Buzzy group, *p* = 0.71). In the Buzzy group, we recorded difficult vascular access in three cases: two were caused by vibration and one by arm stiffness due to cold. In the control group, one child became agitated because they were scared of the procedure. The number of adverse events was not significantly different in the two groups (*p* = 0.36).

DISCUSSION

Children with cognitive impairment suffer significantly more pain than healthy children, with needle-related procedures being the most frequent cause of iatrogenic pain.

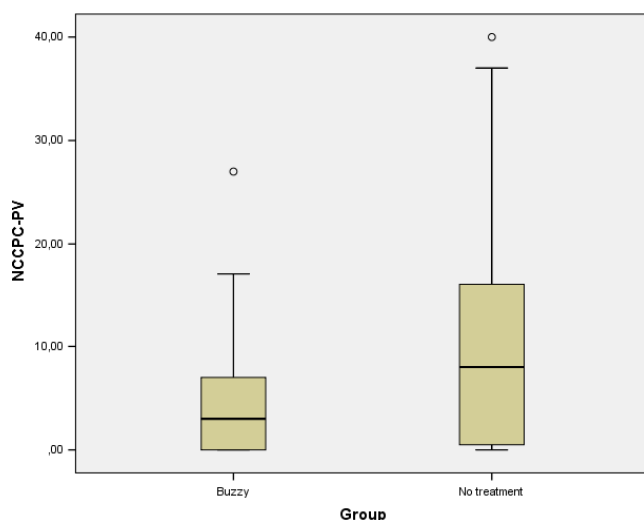


Figure 2 Procedural pain score in the two groups.

Nonpharmacological approaches are the basis for all pain management, but the use of psychological techniques in children with cognitive impairment is limited. Pharmacological analgesia for pain relief during vascular access is the first choice in many hospital settings, but it is not always possible. In this event, physical analgesia should be considered.

This study shows that providing physical analgesia using the Buzzy significantly decreased procedural pain in children with cognitive impairment during vascular access. It was the first study to test the effectiveness of this device in pain relief in children with cognitive impairment.

Parents recorded procedural pain using the NCCPC-PV, with scores ranging from 0 to 80. A pain score of >10.0 indicated the presence of moderate-to-severe pain. The NCCPC-PV has been validated to record pain in children with cognitive impairment during several painful hospital procedures, such as dental extractions, G-button insertions and removal, heel cord tendon lengthening, endoscopies and subcutaneous venous access device insertion (18,19). In these settings, mean procedural pain reported was 12.2, which was slightly higher than in our study: 3.0 in the Buzzy group and 8.0 in the no-intervention group. The mean procedural pain reported in our study was conversely similar to that reported in a recent study that accessed procedural pain exclusively during vascular access (20).

In this study, the Buzzy device was well tolerated by children and the application did not influence the procedural success rate at the first attempt.

This study has some limits. It was not blinded, but blinding was not possible. The reliability of the parents' evaluation of pain before the assessments was not tested. We chose to use the parents' assessment as evidence has shown that the parents of children with cognitive impairment can reliably identify pain in their children (21). Remarkably, many parents in the treatment group asked us to use the device in their children's subsequent procedures.

To respect the ethical standards of the institute, only children who could not receive pharmacological topical anaesthesia, which is the standard treatment in our Institute, were enrolled, thus limiting the selection of patients. Nonetheless, the distribution of patients between the two groups was well balanced.

CONCLUSION

The findings of this study show that children with cognitive impairment benefitted from the analgesia provided by the cooling vibration of the Buzzy during vascular access. Further studies on larger series are needed to confirm these results. The use of vibration and cold analgesia should be considered as part of the procedural pain management of children with cognitive impairment.

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CONFLICTS OF INTEREST

The authors do not have any conflict of interest to declare.

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